

AVR Programming — Balanced Projects

Make: AVR Programming | Chapter 7 → 20 | Progressive Difficulty

■ Ch.7 — Analog-to-Digital Conversion I

PROJECT

Analog Theremin

WHAT TO BUILD

Read a photoresistor on ADC0 using bare metal ADMUX/ADCSRA registers. Map ADC value to a frequency and drive a piezo buzzer using Timer2 in CTC mode — change pitch by waving your hand over the sensor. Add a second potentiometer on ADC1 to control volume by adjusting PWM duty cycle on the buzzer pin. Print current frequency and ADC value to UART. No `analogRead()`, no `tone()` library.

ELECTRONICS

Photoresistor, 10k resistor, potentiometer, piezo buzzer

3D PRINT

Small theremin enclosure — sensor window on top, buzzer grille on front, potentiometer knob hole on side

DONE WHEN

Pitch changes smoothly with hand distance, volume knob works, UART shows correct ADC values and frequency

■ Ch.8 — Hardware Interrupts

PROJECT

Interrupt-Driven Digital Tachometer

WHAT TO BUILD

Attach a Hall effect sensor to INT0. Each magnet pass on a spinning motor triggers an interrupt. Measure time between interrupts using Timer1 counter to calculate RPM. Store last 8 readings in circular buffer, output rolling average RPM via UART every 500ms. Display RPM on 4-digit 7-segment display driven via shift register. Implement high/low RPM threshold alerts — LED changes color. Zero polling anywhere.

ELECTRONICS

Hall effect sensor, magnet, 4-digit 7-segment display, 74HC595 shift register, red and green LEDs

3D PRINT

Tachometer housing — display window on front, sensor mount bracket on side, clip mount for attaching to motor

DONE WHEN

RPM accurate within 2%, circular buffer smooths readings, threshold alerts trigger correctly, UART output at exact 500ms

■ Ch.9 — Timer/Counter Hardware

PROJECT

Precision Stopwatch with Lap Timer

WHAT TO BUILD

Timer1 in CTC mode generates exactly 1ms ticks via ISR. Three buttons: START/STOP, LAP, RESET. Display elapsed time MM:SS:ms on I2C OLED using bare metal I2C registers. Store up to 5 lap times in SRAM. On STOP, show best lap, worst lap, average lap via UART. Implement button debounce in the timer ISR — no delay(). Accuracy must be within 10ms over 60 seconds.

ELECTRONICS

3x push buttons, 3x 10k resistors, SSD1306 OLED display

3D PRINT

Stopwatch enclosure — OLED window on front, 3 button holes on top, belt clip on back

DONE WHEN

Timing accurate within 10ms over 60s, all 5 laps stored correctly, best/worst/average calculated correctly, debounce eliminates bounce

■ Ch.10 — Pulse-Width Modulation

PROJECT

RGB LED Mood Lamp with Sensor Reactions

WHAT TO BUILD

Three Timer PWM channels control R, G, B — direct register access only, no analogWrite(). Implement smooth HSV color transitions in fixed-point integer math. Four modes triggered by button: rainbow cycle, breathing, temperature reactive (read thermistor on ADC, hot=red cold=blue), sound reactive (microphone on ADC, loud=bright). Potentiometer controls transition speed. Mode and speed saved to EEPROM on power down.

ELECTRONICS

RGB LED common cathode, 3x 220Ω resistors, potentiometer, push button, thermistor, microphone module

3D PRINT

Frosted dome lamp — print dome in white or translucent PETG for diffusion, base holds all electronics, mode button on top of base

DONE WHEN

Smooth transitions no flicker, all 4 modes work correctly, temperature and sound reactive modes respond in real time, settings survive power cycle

■ Ch.11 — Driving Servo Motors

PROJECT

UART-Controlled Pan-Tilt Camera Mount

WHAT TO BUILD

Generate precise 50Hz PWM for 2 servos using Timer1 bare metal — no Servo library. UART interface accepts: PAN:90, TILT:45, CENTER, SWEEP PAN, SWEEP TILT. Implement smooth interpolation — move from current to target angle in 2-degree steps with 20ms delay between steps via timer ISR. Display current angles on OLED. Add limit switches on INT0/INT1 — servo stops and reverses at physical limits. Save home position to EEPROM.

ELECTRONICS

2x MG996R servos, 2x limit switches, SSD1306 OLED, external 5V supply

3D PRINT

Pan-tilt bracket — two axis mount with servo pockets, limit switch mounts, camera plate on top, cable routing through arm

DONE WHEN

Smooth movement between positions, limits trigger correctly, UART commands work reliably, home position loads from EEPROM on boot

■ Ch.12 — Analog-to-Digital Conversion II

PROJECT

4-Channel Analog Sensor Dashboard

WHAT TO BUILD

Free-running ADC with interrupt cycles through 4 channels: temperature (thermistor), light (LDR), sound (microphone), and potentiometer. Apply moving average filter to each channel — 8 sample window. Display all 4 channels as live bar graphs on SSD1306 OLED. UART outputs CSV data at 5Hz for logging. Button cycles through display modes: all channels, single channel detail, min/max history. Store min/max per channel in EEPROM.

ELECTRONICS

Thermistor, LDR, microphone module, potentiometer, 4x 10k resistors, SSD1306 OLED

3D PRINT

Desktop sensor station — 4 sensor windows on top panel, OLED display on front face, USB connector for power, angled stand for easy viewing

DONE WHEN

All 4 channels update smoothly, moving average reduces noise visibly, bar graphs scale correctly, CSV logs correctly, EEPROM min/max survives power cycle

■ Ch.13 — Advanced PWM Tricks

PROJECT

Servo Sine Wave Animator

WHAT TO BUILD

Store one full sine wave cycle as 64 values in PROGMEM. Timer2 ISR fires at 64Hz, reads next PROGMEM value, outputs to 3 servos with 120-degree phase offset each — creating a smooth wave motion like a snake or walking mechanism. Potentiometer on ADC controls animation speed 0.5Hz to 5Hz. Second potentiometer controls amplitude. Button freezes animation at current position. Display speed and amplitude on UART. This is the foundation of multi-joint coordinated motion.

ELECTRONICS

3x servos, 2x potentiometers, push button, external 5V supply

3D PRINT

3-segment snake mechanism — each segment connects to next via servo horn, undulating motion when animation runs, printable in flexible TPU optional

DONE WHEN

Smooth sine wave motion, 120-degree phase offset visible, speed and amplitude controls work in real time, freeze button works cleanly

■ Ch.14 — Switches

PROJECT

4-Button Menu System with OLED

WHAT TO BUILD

Build a reusable button driver: single press, double press, long hold — timer-based debounce, no delay(). Use driver to implement a full settings menu on SSD1306 OLED: navigate with UP/DOWN, SELECT to enter submenu, BACK to exit. Menu tree: Display Settings (brightness, contrast), Sensor Settings (sample rate, filter size), System (reset, about). Store all settings in EEPROM. Apply settings immediately when changed. This driver becomes reusable across all future projects.

ELECTRONICS

4x push buttons, 4x 10k resistors, SSD1306 OLED

3D PRINT

Control panel — OLED window, 4 labeled button holes, stands at 30-degree angle for easy viewing, USB power on back

DONE WHEN

Single/double/hold all reliable, full menu tree navigable, settings apply immediately, all settings persist in EEPROM, driver packaged as reusable .h/.c files

■ Ch.15 — Advanced Motors

PROJECT

Stepper Motor Plotter

WHAT TO BUILD

Two NEMA17 steppers drive X and Y axes. Timer1 ISR generates step pulses. Implement Bresenham line algorithm for coordinated movement — both axes start and stop together. Simple acceleration ramp in ISR — no jerky starts. UART accepts commands: LINE X:100 Y:50, HOME, SPEED:500. Draw shapes: square, triangle, circle (approximated with 36 line segments). Limit switches home on startup. Track absolute position in steps. A pen holder completes a basic drawing machine.

ELECTRONICS

2x NEMA17, 2x A4988, 4x limit switches, 12V PSU, 100µF capacitor

3D PRINT

2-axis plotter frame — CoreXY or cartesian design, motor mounts, belt tensioners, limit switch brackets, pen holder with spring loaded tip

DONE WHEN

Lines straight on logic analyzer, diagonal moves coordinated, homing repeatable, all shapes drawn correctly, acceleration smooth

■ Ch.16 — SPI

PROJECT

Bare Metal SPI Display with Live Gauges

WHAT TO BUILD

Drive GC9A01A round display using raw SPCR/SPDR registers — your own spi_init(), spi_transfer(), spi_cs() functions. Read 3 ADC channels: temperature, light, potentiometer. Render on display: analog gauge needle for temperature, brightness bar for light, large number for pot value. Draw gauge faces using sin/cos lookup table in PROGMEM. Verify SPI signals on logic analyzer — check clock polarity, phase, CS timing. Target 15fps minimum.

ELECTRONICS

GC9A01A display, thermistor, LDR, potentiometer, 3x 10k resistors

3D PRINT

Round gauge display mount — bezel fits GC9A01A exactly, desk stand at 45-degree angle, sensor inputs through base

DONE WHEN

Display initializes bare metal, gauges update smoothly at 15fps, SPI verified on logic analyzer, all 3 sensors reflected correctly

■ Ch.17 — I2C

PROJECT

Bare Metal MPU6050 Tilt Display

WHAT TO BUILD

Write I2C driver from scratch using TWCR/TWDR/TWSR registers — i2c_start(), i2c_stop(), i2c_write(), i2c_read(). Read MPU6050 accelerometer at 100Hz. Implement complementary filter combining accelerometer and gyroscope for stable pitch and roll angles. Display on GC9A01A: artificial horizon — a line that tilts with the board, roll and pitch numbers. UART outputs roll, pitch at 10Hz CSV. Verify I2C transactions on logic analyzer — ACK bits, address phase.

ELECTRONICS

MPU6050, GC9A01A display, 2x 4.7k pull-up resistors

3D PRINT

IMU display unit — MPU6050 mount at exact center, display bezel, stand that lets you tilt the whole unit to see horizon move

DONE WHEN

I2C verified on logic analyzer, pitch and roll stable within 2 degrees at rest, horizon updates smoothly, UART CSV logs correctly

■ Ch.18 — Using Flash Program Memory

PROJECT

PROGMEM Animation Player

WHAT TO BUILD

Store 8 animation frames for WS2812B 8x8 LED matrix in PROGMEM. Store 4 complete animations: fire effect, scrolling text, spinning shape, custom pattern you design. Animation player reads frames with `pgm_read_byte()`. Timer1 controls frame rate. Button cycles animations. Potentiometer controls speed. Measure and display SRAM usage with and without PROGMEM via UART — show exactly how many bytes you saved. Second button triggers single-frame step mode for debugging.

ELECTRONICS

WS2812B 8x8 matrix, 470Ω resistor, 1000μF capacitor, 2x buttons, potentiometer

3D PRINT

Display frame with diffuser — border surround for matrix, frosted PETG diffuser layer, tilted stand, button and knob holes on base

DONE WHEN

All 4 animations play correctly, SRAM savings confirmed in UART output, frame rate consistent, step mode works for debugging

■ Ch.19 — EEPROM

PROJECT

Self-Calibrating Sensor Node

WHAT TO BUILD

Sensor node reads LDR and thermistor. On first boot runs auto-calibration: collect 100 samples, compute min/max/offset, store in EEPROM with CRC8 checksum. On subsequent boots: load calibration, verify checksum, use values. UART interface: CALIBRATE, SHOW CAL, RESET CAL, READ. Apply calibration to produce accurate lux and temperature values. Display on OLED with calibrated units. Boot counter in EEPROM. Alert if checksum fails — red LED and UART warning.

ELECTRONICS

LDR, thermistor, 2x 10k resistors, SSD1306 OLED, red LED

3D PRINT

Sensor enclosure — vented for thermistor, LDR window, OLED display, wall mountable, USB power

DONE WHEN

Calibration survives power cycle, CRC detects corruption, calibrated readings accurate, boot counter increments, UART commands all work

■ Ch.20 — Final Integration Project

PROJECT

Bare Metal Weather Station and Display

WHAT TO BUILD

Combine everything into one polished system. Bare metal I2C reads MPU6050 (tilt/vibration detection) and BMP280 (temperature/pressure/altitude). Bare metal SPI drives GC9A01A showing: temperature gauge, pressure bar, altitude number, tilt indicator. WS2812B LED ring shows weather condition color — blue cold, red hot, green normal. EEPROM stores min/max records and calibration. PROGMEM stores gauge face graphics and font. ADC reads battery voltage. Button cycles display screens. UART outputs all data as CSV at 1Hz. Everything interrupt driven, no delay() anywhere.

ELECTRONICS

MPU6050, BMP280, GC9A01A display, WS2812B LED ring, SSD1306 OLED, push button, voltage divider for battery

3D PRINT

Weather station enclosure — round GC9A01A on front, LED ring around display bezel, sensor vents on sides, OLED on top, wall mount bracket on back, full PETG build

DONE WHEN

All sensors read correctly, display updates at 10fps, LED ring color matches temperature, EEPROM records survive power cycle, UART CSV logs all fields, zero delay() in codebase
